

sg13g2_io_typ_1p5V_3p3V_25C Library

Cell Groups
INOUTx
INPUT
SG13G2_CORNER
SG13G2_FILLER10000
SG13G2_FILLER1000
SG13G2_FILLER2000
SG13G2_FILLER200
SG13G2_FILLER4000
SG13G2_FILLER400
SG13G2_IOPADANALOG
SG13G2_IOPADIOVDD
SG13G2_IOPADIOVSS
SG13G2_IOPADVDD
SG13G2_IOPADVSS
TRI_OUTx

INOUTx



*sg13g2_io_typ_1p5V_3p3V_25C Cell Library: Process
sg13g2_io_typ_1p5V_3p3V_25C, Voltage 1.50, Temp 25.00*

Truth Table

INPUT			OUTPUT	
c2p	c2p_en	pad	pad	p2c
-	0	0	-	0
-	0	1	-	1
0	1	-	0	0
1	1	-	1	1

Footprint

Cell Name	Area
sg13g2_IOPadInOut16mA	14400.00000
sg13g2_IOPadInOut30mA	14400.00000
sg13g2_IOPadInOut4mA	14400.00000
sg13g2_IOPadOut16mA	14400.00000
sg13g2_IOPadOut30mA	14400.00000
sg13g2_IOPadOut4mA	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	c2p	c2p_en	pad	p2c	pad
sg13g2_IOPadInOut16mA	0.02935	0.02715	0.29651	0.24000	20.00000
sg13g2_IOPadInOut30mA	0.02935	0.02715	0.38418	0.24000	40.00000
sg13g2_IOPadInOut4mA	0.02935	0.02714	0.22079	0.24000	12.00000
sg13g2_IOPadOut16mA	0.04118	0.00000	0.00000	0.00000	20.00000
sg13g2_IOPadOut30mA	0.04119	0.00000	0.00000	0.00000	40.00000
sg13g2_IOPadOut4mA	0.04119	0.00000	0.00000	0.00000	12.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadInOut16mA	48.97630	2959.84877	6207.43000
sg13g2_IOPadInOut30mA	152.30200	3102.58894	6207.33000
sg13g2_IOPadInOut4mA	65.09600	2896.16879	6206.91000
sg13g2_IOPadOut16mA	265.92300	890.73050	1933.24000
sg13g2_IOPadOut30mA	463.62200	1039.03233	1932.46000
sg13g2_IOPadOut4mA	181.32800	802.66250	1933.82000

Delay Information

Delay(ns) to p2c rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad->p2c (RR)	0.12000	0.02400	0.21024	0.80000	0.14400	0.36797	5.00000	0.24000	0.62755
sg13g2_IOPadInOut30mA	pad->p2c (RR)	0.12000	0.02400	0.20958	0.80000	0.14400	0.36820	5.00000	0.24000	0.62795
sg13g2_IOPadInOut4mA	pad->p2c (RR)	0.12000	0.02400	0.21028	0.80000	0.14400	0.36794	5.00000	0.24000	0.62591

Delay(ns) to p2c falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad->p2c (FF)	0.12000	0.02400	0.21293	0.80000	0.14400	0.37776	5.00000	0.24000	0.69199
sg13g2_IOPadInOut30mA	pad->p2c (FF)	0.12000	0.02400	0.21295	0.80000	0.14400	0.37793	5.00000	0.24000	0.69216
sg13g2_IOPadInOut4mA	pad->p2c (FF)	0.12000	0.02400	0.21282	0.80000	0.14400	0.37753	5.00000	0.24000	0.69177

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p->pad (RR)	0.02000	1.00000	1.44327	0.33000	9.00000	2.44766	2.50000	20.00000	3.65879
	c2p_en->pad (FR)	0.02000	1.00000	1.02938	0.33000	9.00000	1.07919	2.50000	20.00000	1.24399
	c2p_en->pad (RR)	0.02000	1.00000	1.44074	0.33000	9.00000	2.50092	2.50000	20.00000	3.79243
sg13g2_IOPadInOut30mA	c2p->pad (RR)	0.02000	2.00000	1.78898	0.33000	18.00000	3.01065	2.50000	40.00000	4.32202
	c2p_en->pad (FR)	0.02000	2.00000	1.22138	0.33000	18.00000	1.27289	2.50000	40.00000	1.42994
	c2p_en->pad (RR)	0.02000	2.00000	1.78543	0.33000	18.00000	3.07252	2.50000	40.00000	4.47953
sg13g2_IOPadInOut4mA	c2p->pad (RR)	0.02000	1.00000	1.46628	0.33000	5.00000	3.22961	2.50000	12.00000	6.27101
	c2p_en->pad (FR)	0.02000	1.00000	0.83945	0.33000	5.00000	0.89044	2.50000	12.00000	1.04949
	c2p_en->pad (RR)	0.02000	1.00000	1.47499	0.33000	5.00000	3.30926	2.50000	12.00000	6.48105
sg13g2_IOPadOut16mA	c2p->pad (RR)	0.02000	1.00000	1.39415	0.33000	9.00000	2.47884	2.50000	20.00000	4.29533
sg13g2_IOPadOut30mA	c2p->pad (RR)	0.02000	2.00000	1.74419	0.33000	18.00000	3.04502	2.50000	40.00000	4.96326
sg13g2_IOPadOut4mA	c2p->pad (RR)	0.02000	1.00000	1.40850	0.33000	5.00000	3.25046	2.50000	12.00000	6.88983

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p->pad (FF)	0.02000	1.00000	1.02629	0.33000	9.00000	1.93793	2.50000	20.00000	3.35313
	c2p_en->pad (FF)	0.02000	1.00000	1.01328	0.33000	9.00000	1.08131	2.50000	20.00000	1.30466
	c2p_en->pad (RF)	0.02000	1.00000	0.85844	0.33000	9.00000	1.86664	2.50000	20.00000	3.14182
sg13g2_IOPadInOut30mA	c2p->pad (FF)	0.02000	2.00000	1.39091	0.33000	18.00000	2.41639	2.50000	40.00000	3.86934
	c2p_en->pad (FF)	0.02000	2.00000	1.60342	0.33000	18.00000	1.66822	2.50000	40.00000	1.86602
	c2p_en->pad (RF)	0.02000	2.00000	1.03300	0.33000	18.00000	2.18655	2.50000	40.00000	3.54670
sg13g2_IOPadInOut4mA	c2p->pad (FF)	0.02000	1.00000	1.09273	0.33000	5.00000	2.86282	2.50000	12.00000	6.12841
	c2p_en->pad (FF)	0.02000	1.00000	0.51623	0.33000	5.00000	0.58380	2.50000	12.00000	0.81186
	c2p_en->pad (RF)	0.02000	1.00000	1.11274	0.33000	5.00000	2.99826	2.50000	12.00000	6.25266
sg13g2_IOPadOut16mA	c2p->pad (FF)	0.02000	1.00000	0.96705	0.33000	9.00000	1.85066	2.50000	20.00000	3.11929
sg13g2_IOPadOut30mA	c2p->pad (FF)	0.02000	2.00000	1.33427	0.33000	18.00000	2.33748	2.50000	40.00000	3.64438
sg13g2_IOPadOut4mA	c2p->pad (FF)	0.02000	1.00000	1.01204	0.33000	5.00000	2.76105	2.50000	12.00000	5.87621

Power Information

Internal switching power(pJ) to p2c rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad	0.12000	0.02400	-1.27772	0.80000	0.14400	-1.16179	5.00000	0.24000	-0.33306
	pad	0.12000	0.02400	0.03877	0.80000	0.14400	0.04018	5.00000	0.24000	0.04249
sg13g2_IOPadInOut30mA	pad	0.12000	0.02400	-2.00018	0.80000	0.14400	-1.87673	5.00000	0.24000	-1.04877
	pad	0.12000	0.02400	0.03892	0.80000	0.14400	0.04061	5.00000	0.24000	0.04282
sg13g2_IOPadInOut4mA	pad	0.12000	0.02400	-0.69464	0.80000	0.14400	-0.58134	5.00000	0.24000	0.24982
	pad	0.12000	0.02400	0.03873	0.80000	0.14400	0.04009	5.00000	0.24000	0.04194

Internal switching power(pJ) to p2c falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad	0.12000	0.02400	1.74503	0.80000	0.14400	1.85172	5.00000	0.24000	2.68707
	pad	0.12000	0.02400	0.10225	0.80000	0.14400	0.11221	5.00000	0.24000	0.15551
sg13g2_IOPadInOut30mA	pad	0.12000	0.02400	2.46373	0.80000	0.14400	2.56472	5.00000	0.24000	3.41589
	pad	0.12000	0.02400	0.10200	0.80000	0.14400	0.11187	5.00000	0.24000	0.15518
sg13g2_IOPadInOut4mA	pad	0.12000	0.02400	1.16046	0.80000	0.14400	1.27269	5.00000	0.24000	2.10755
	pad	0.12000	0.02400	0.10244	0.80000	0.14400	0.11247	5.00000	0.24000	0.15574

Internal switching power(pJ) to pad rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p	0.02000	1.00000	6.56520	0.33000	9.00000	6.43704	2.50000	20.00000	6.08398
	c2p	0.02000	1.00000	2.33571	0.33000	9.00000	20.09920	2.50000	20.00000	37.33310
	c2p_en	0.02000	1.00000	6.23838	0.33000	9.00000	8.69236	2.50000	20.00000	11.67040
	c2p_en	0.02000	1.00000	2.28973	0.33000	9.00000	20.06930	2.50000	20.00000	37.06230
sg13g2_IOPadInOut30mA	c2p	0.02000	2.00000	10.13520	0.33000	18.00000	9.49426	2.50000	40.00000	8.62884
	c2p	0.02000	2.00000	4.55732	0.33000	18.00000	39.25900	2.50000	40.00000	20.05650
	c2p_en	0.02000	2.00000	9.58132	0.33000	18.00000	14.50480	2.50000	40.00000	21.39890
	c2p_en	0.02000	2.00000	4.51195	0.33000	18.00000	39.27120	2.50000	40.00000	19.93290
sg13g2_IOPadInOut4mA	c2p	0.02000	1.00000	3.78504	0.33000	5.00000	4.13523	2.50000	12.00000	5.25205
	c2p	0.02000	1.00000	2.34279	0.33000	5.00000	11.25430	2.50000	12.00000	27.30610
	c2p_en	0.02000	1.00000	3.56789	0.33000	5.00000	4.79140	2.50000	12.00000	7.38708
	c2p_en	0.02000	1.00000	2.29661	0.33000	5.00000	11.21740	2.50000	12.00000	27.33740
sg13g2_IOPadOut16mA	c2p	0.02000	1.00000	6.13545	0.33000	9.00000	5.98089	2.50000	20.00000	5.78487
	c2p	0.02000	1.00000	-0.02783	0.33000	9.00000	0.00518	2.50000	20.00000	0.32671
sg13g2_IOPadOut30mA	c2p	0.02000	2.00000	9.84264	0.33000	18.00000	9.34969	2.50000	40.00000	8.75259
	c2p	0.02000	2.00000	-0.02783	0.33000	18.00000	0.00518	2.50000	40.00000	0.33211
sg13g2_IOPadOut4mA	c2p	0.02000	1.00000	3.19107	0.33000	5.00000	3.09689	2.50000	12.00000	3.03758
	c2p	0.02000	1.00000	-0.02783	0.33000	5.00000	0.00519	2.50000	12.00000	0.33268

Internal switching power(pJ) to pad falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p	0.02000	1.00000	14.87970	0.33000	9.00000	9.84385	2.50000	20.00000	8.27807
	c2p	0.02000	1.00000	0.29534	0.33000	9.00000	0.33742	2.50000	20.00000	0.68740
	c2p_en	0.02000	1.00000	3.56961	0.33000	9.00000	3.83939	2.50000	20.00000	4.24220
	c2p_en	0.02000	1.00000	0.15866	0.33000	9.00000	0.19342	2.50000	20.00000	0.39711
sg13g2_IOPadInOut30mA	c2p	0.02000	2.00000	53.32300	0.33000	18.00000	32.52670	2.50000	40.00000	23.97180
	c2p	0.02000	2.00000	0.29870	0.33000	18.00000	0.33845	2.50000	40.00000	0.67745
	c2p_en	0.02000	2.00000	5.40792	0.33000	18.00000	5.68301	2.50000	40.00000	6.10851
	c2p_en	0.02000	2.00000	0.15860	0.33000	18.00000	0.19457	2.50000	40.00000	0.39003
sg13g2_IOPadInOut4mA	c2p	0.02000	1.00000	2.75608	0.33000	5.00000	3.32942	2.50000	12.00000	4.37140
	c2p	0.02000	1.00000	0.30373	0.33000	5.00000	0.35861	2.50000	12.00000	0.72902
	c2p_en	0.02000	1.00000	2.14560	0.33000	5.00000	2.73094	2.50000	12.00000	3.76638
	c2p_en	0.02000	1.00000	0.16702	0.33000	5.00000	0.21430	2.50000	12.00000	0.44017
sg13g2_IOPadOut16mA	c2p	0.02000	1.00000	15.52820	0.33000	9.00000	9.48116	2.50000	20.00000	7.61589
	c2p	0.02000	1.00000	0.09591	0.33000	9.00000	0.13472	2.50000	20.00000	0.45989
sg13g2_IOPadOut30mA	c2p	0.02000	2.00000	55.03470	0.33000	18.00000	32.66230	2.50000	40.00000	24.05830
	c2p	0.02000	2.00000	0.09591	0.33000	18.00000	0.13400	2.50000	40.00000	0.46037
sg13g2_IOPadOut4mA	c2p	0.02000	1.00000	2.33191	0.33000	5.00000	2.25783	2.50000	12.00000	2.24360
	c2p	0.02000	1.00000	0.09589	0.33000	5.00000	0.13477	2.50000	12.00000	0.46231

Passive power(pJ) for c2p rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	-0.00010	0.33000	-0.00008	2.50000	-0.00008
	0.02000	-0.02940	0.33000	-0.03065	2.50000	-0.03152
sg13g2_IOPadInOut30mA	0.02000	-0.00011	0.33000	-0.00008	2.50000	-0.00008
	0.02000	-0.02940	0.33000	-0.03065	2.50000	-0.03152
sg13g2_IOPadInOut4mA	0.02000	-0.00010	0.33000	-0.00008	2.50000	-0.00008
	0.02000	-0.02943	0.33000	-0.03065	2.50000	-0.03152

Passive power(pJ) for c2p falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	0.00010	0.33000	0.00008	2.50000	0.00008
	0.02000	0.03834	0.33000	0.03778	2.50000	0.03769
sg13g2_IOPadInOut30mA	0.02000	0.00011	0.33000	0.00008	2.50000	0.00008
	0.02000	0.03834	0.33000	0.03778	2.50000	0.03769
sg13g2_IOPadInOut4mA	0.02000	0.00010	0.33000	0.00008	2.50000	0.00008
	0.02000	0.03834	0.33000	0.03778	2.50000	0.03769

Passive power(pJ) for c2p rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	(!c2p_en * pad * p2c)	0.02000	-0.00010	0.33000	-0.00008	2.50000	-0.00008
	(!c2p_en * pad * p2c)	0.02000	-0.02940	0.33000	-0.03065	2.50000	-0.03152
	(!c2p_en * !pad * !p2c)	0.02000	-0.00010	0.33000	-0.00008	2.50000	-0.00008
	(!c2p_en * !pad * !p2c)	0.02000	-0.02940	0.33000	-0.03065	2.50000	-0.03152
sg13g2_IOPadInOut30mA	(!c2p_en * pad * p2c)	0.02000	-0.00010	0.33000	-0.00008	2.50000	-0.00008
	(!c2p_en * pad * p2c)	0.02000	-0.02940	0.33000	-0.03065	2.50000	-0.03152
	(!c2p_en * !pad * !p2c)	0.02000	-0.00011	0.33000	-0.00008	2.50000	-0.00008
	(!c2p_en * !pad * !p2c)	0.02000	-0.02940	0.33000	-0.03065	2.50000	-0.03152
sg13g2_IOPadInOut4mA	(!c2p_en * pad * p2c)	0.02000	-0.00010	0.33000	-0.00008	2.50000	-0.00008
	(!c2p_en * pad * p2c)	0.02000	-0.02943	0.33000	-0.03065	2.50000	-0.03152
	(!c2p_en * !pad * !p2c)	0.02000	-0.00010	0.33000	-0.00008	2.50000	-0.00008
	(!c2p_en * !pad * !p2c)	0.02000	-0.02943	0.33000	-0.03065	2.50000	-0.03152

Passive power(pJ) for c2p falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	(!c2p_en * pad * p2c)	0.02000	0.00010	0.33000	0.00008	2.50000	0.00008
	(!c2p_en * pad * p2c)	0.02000	0.03834	0.33000	0.03778	2.50000	0.03769
	(!c2p_en * !pad * !p2c)	0.02000	0.00010	0.33000	0.00008	2.50000	0.00008
	(!c2p_en * !pad * !p2c)	0.02000	0.03834	0.33000	0.03778	2.50000	0.03769
sg13g2_IOPadInOut30mA	(!c2p_en * pad * p2c)	0.02000	0.00010	0.33000	0.00008	2.50000	0.00008
	(!c2p_en * pad * p2c)	0.02000	0.03834	0.33000	0.03778	2.50000	0.03769
	(!c2p_en * !pad * !p2c)	0.02000	0.00011	0.33000	0.00008	2.50000	0.00008
	(!c2p_en * !pad * !p2c)	0.02000	0.03834	0.33000	0.03778	2.50000	0.03769
sg13g2_IOPadInOut4mA	(!c2p_en * pad * p2c)	0.02000	0.00010	0.33000	0.00008	2.50000	0.00008
	(!c2p_en * pad * p2c)	0.02000	0.03834	0.33000	0.03778	2.50000	0.03769
	(!c2p_en * !pad * !p2c)	0.02000	0.00010	0.33000	0.00008	2.50000	0.00008
	(!c2p_en * !pad * !p2c)	0.02000	0.03834	0.33000	0.03778	2.50000	0.03769

Passive power(pJ) for c2p_en rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	1.57457	0.33000	1.57490	2.50000	1.57832
	0.02000	0.05056	0.33000	0.06662	2.50000	0.25452
sg13g2_IOPadInOut30mA	0.02000	2.53315	0.33000	2.53231	2.50000	2.55000
	0.02000	0.05055	0.33000	0.06667	2.50000	0.25434
sg13g2_IOPadInOut4mA	0.02000	0.75555	0.33000	0.75573	2.50000	0.75628
	0.02000	0.05057	0.33000	0.06663	2.50000	0.25432

Passive power(pJ) for c2p_en falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	0.34468	0.33000	0.34274	2.50000	0.33981
	0.02000	0.13337	0.33000	0.15158	2.50000	0.34168
sg13g2_IOPadInOut30mA	0.02000	0.33563	0.33000	0.33373	2.50000	0.33071
	0.02000	0.13339	0.33000	0.15160	2.50000	0.34178
sg13g2_IOPadInOut4mA	0.02000	0.36761	0.33000	0.36645	2.50000	0.36273
	0.02000	0.13335	0.33000	0.15157	2.50000	0.34186

Passive power(pJ) for pad rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	6.56520	0.33000	6.43704	2.50000	6.08398
	0.02000	2.33571	0.33000	20.09920	2.50000	37.33310
	0.02000	6.23838	0.33000	8.69236	2.50000	11.67040
	0.02000	2.28973	0.33000	20.06930	2.50000	37.06230
sg13g2_IOPadInOut30mA	0.02000	10.13520	0.33000	9.49426	2.50000	8.62884
	0.02000	4.55732	0.33000	39.25900	2.50000	20.05650
	0.02000	9.58132	0.33000	14.50480	2.50000	21.39890
	0.02000	4.51195	0.33000	39.27120	2.50000	19.93290
sg13g2_IOPadInOut4mA	0.02000	3.78504	0.33000	4.13523	2.50000	5.25205
	0.02000	2.34279	0.33000	11.25430	2.50000	27.30610
	0.02000	3.56789	0.33000	4.79140	2.50000	7.38708
	0.02000	2.29661	0.33000	11.21740	2.50000	27.33740
sg13g2_IOPadOut16mA	0.02000	6.13545	0.33000	5.98089	2.50000	5.78487
	0.02000	-0.02783	0.33000	0.00518	2.50000	0.32671
sg13g2_IOPadOut30mA	0.02000	9.84264	0.33000	9.34969	2.50000	8.75259
	0.02000	-0.02783	0.33000	0.00518	2.50000	0.33211
sg13g2_IOPadOut4mA	0.02000	3.19107	0.33000	3.09689	2.50000	3.03758
	0.02000	-0.02783	0.33000	0.00519	2.50000	0.33268

Passive power(pJ) for pad falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	14.87970	0.33000	9.84385	2.50000	8.27807
	0.02000	0.29534	0.33000	0.33742	2.50000	0.68740
	0.02000	3.56961	0.33000	3.83939	2.50000	4.24220
	0.02000	0.15866	0.33000	0.19342	2.50000	0.39711
sg13g2_IOPadInOut30mA	0.02000	53.32300	0.33000	32.52670	2.50000	23.97180
	0.02000	0.29870	0.33000	0.33845	2.50000	0.67745
	0.02000	5.40792	0.33000	5.68301	2.50000	6.10851
	0.02000	0.15860	0.33000	0.19457	2.50000	0.39003
sg13g2_IOPadInOut4mA	0.02000	2.75608	0.33000	3.32942	2.50000	4.37140
	0.02000	0.30373	0.33000	0.35861	2.50000	0.72902
	0.02000	2.14560	0.33000	2.73094	2.50000	3.76638
	0.02000	0.16702	0.33000	0.21430	2.50000	0.44017
sg13g2_IOPadOut16mA	0.02000	15.52820	0.33000	9.48116	2.50000	7.61589
	0.02000	0.09591	0.33000	0.13472	2.50000	0.45989
sg13g2_IOPadOut30mA	0.02000	55.03470	0.33000	32.66230	2.50000	24.05830
	0.02000	0.09591	0.33000	0.13400	2.50000	0.46037
sg13g2_IOPadOut4mA	0.02000	2.33191	0.33000	2.25783	2.50000	2.24360
	0.02000	0.09589	0.33000	0.13477	2.50000	0.46231

INPUT



*sg13g2_io_typ_1p5V_3p3V_25C Cell Library: Process
sg13g2_io_typ_1p5V_3p3V_25C, Voltage 1.50, Temp 25.00*

Truth Table

INPUT	OUTPUT
pad	p2c
0	0
1	1

Footprint

Cell Name	Area
sg13g2_IOPadIn	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	pad	p2c
sg13g2_IOPadIn	0.23445	0.24000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadIn	108.10100	407.60400	974.10500

Delay Information

Delay(ns) to p2c rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad->p2c (RR)	0.12000	0.02400	0.20994	0.80000	0.14400	0.36750	5.00000	0.24000	0.62547

Delay(ns) to p2c falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad->p2c (FF)	0.12000	0.02400	0.21267	0.80000	0.14400	0.37718	5.00000	0.24000	0.69022

Power Information

Internal switching power(pJ) to p2c rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad	0.12000	0.02400	-0.60827	0.80000	0.14400	-0.49395	5.00000	0.24000	0.33095
	pad	0.12000	0.02400	0.03938	0.80000	0.14400	0.04091	5.00000	0.24000	0.04287

Internal switching power(pJ) to p2c falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad	0.12000	0.02400	1.07486	0.80000	0.14400	1.18648	5.00000	0.24000	2.02402
	pad	0.12000	0.02400	0.09788	0.80000	0.14400	0.10791	5.00000	0.24000	0.15075

SG13G2_CORNER



sg13g2_io_typ_1p5V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p5V_3p3V_25C,
Voltage 1.50, Temp 25.00

Footprint

Cell Name	Area
sg13g2_Corner	32400.00000

Pin Capacitance Information

SG13G2_FILLER10000



sg13g2_io_typ_1p5V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p5V_3p3V_25C,
Voltage 1.50, Temp 25.00

Footprint

Cell Name	Area
sg13g2_Filler10000	18000.00000

Pin Capacitance Information

SG13G2_FILLER1000



sg13g2_io_typ_1p5V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p5V_3p3V_25C,
Voltage 1.50, Temp 25.00

Footprint

Cell Name	Area
sg13g2_Filler1000	900.00000

Pin Capacitance Information

SG13G2_FILLER2000



sg13g2_io_typ_1p5V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p5V_3p3V_25C,
Voltage 1.50, Temp 25.00

Footprint

Cell Name	Area
sg13g2_Filler2000	1800.00000

Pin Capacitance Information

SG13G2_FILLER200



sg13g2_io_typ_1p5V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p5V_3p3V_25C,
Voltage 1.50, Temp 25.00

Footprint

Cell Name	Area
sg13g2_Filler200	180.00000

Pin Capacitance Information

SG13G2_FILLER4000



sg13g2_io_typ_1p5V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p5V_3p3V_25C,
Voltage 1.50, Temp 25.00

Footprint

Cell Name	Area
sg13g2_Filler4000	3600.00000

Pin Capacitance Information

SG13G2_FILLER400



sg13g2_io_typ_1p5V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p5V_3p3V_25C,
Voltage 1.50, Temp 25.00

Footprint

Cell Name	Area
sg13g2_Filler400	360.00000

Pin Capacitance Information

SG13G2_IOPADANALOG



sg13g2_io_typ_1p5V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p5V_3p3V_25C,
Voltage 1.50, Temp 25.00

Truth Table

INPUT		OUTPUT	
pad	padres	pad	padres
x	x	-	-

Footprint

Cell Name	Area
sg13g2_IOPadAnalog	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	pad	padres	pad	padres
sg13g2_IOPadAnalog	0.76624	0.60477	30000.00000	30000.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadAnalog	0.00000	517.45000	1034.90000

Delay Information

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadAnalog	pad->pad (R)	10.00000	500.00000	1000.00000	200.00000	30000.00000	1000.00000	200.00000	30000.00000	1000.00000

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadAnalog	pad->pad (F)	10.00000	500.00000	1000.00000	200.00000	30000.00000	1000.00000	200.00000	30000.00000	1000.00000

SG13G2_IOPADIOVDD



sg13g2_io_typ_1p5V_3p3V_25C

Cell Library: Process

sg13g2_io_typ_1p5V_3p3V_25C,

Voltage 1.50, Temp 25.00

Footprint

Cell Name	Area
sg13g2_IOPadIOVdd	14400.00000

Pin Capacitance Information

SG13G2_IOPADIOVSS



sg13g2_io_typ_1p5V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p5V_3p3V_25C,
Voltage 1.50, Temp 25.00

Footprint

Cell Name	Area
sg13g2_IOPadIOVss	14400.00000

Pin Capacitance Information

SG13G2_IOPADVDD



sg13g2_io_typ_1p5V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p5V_3p3V_25C,
Voltage 1.50, Temp 25.00

Footprint

Cell Name	Area
sg13g2_IOPadVdd	14400.00000

Pin Capacitance Information

SG13G2_IOPADVSS



sg13g2_io_typ_1p5V_3p3V_25C
Cell Library: Process
sg13g2_io_typ_1p5V_3p3V_25C,
Voltage 1.50, Temp 25.00

Footprint

Cell Name	Area
sg13g2_IOPadVss	14400.00000

Pin Capacitance Information

TRI_OUTx



*sg13g2_io_typ_1p5V_3p3V_25C Cell Library: Process
sg13g2_io_typ_1p5V_3p3V_25C, Voltage 1.50, Temp
25.00*

Truth Table

INPUT		OUTPUT
c2p	c2p_en	pad
-	0	HiZ
0	1	0
1	1	1

Footprint

Cell Name	Area
sg13g2_IOPadTriOut16mA	14400.00000
sg13g2_IOPadTriOut30mA	14400.00000
sg13g2_IOPadTriOut4mA	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	c2p	c2p_en	pad
sg13g2_IOPadTriOut16mA	0.02883	0.02771	20.00000
sg13g2_IOPadTriOut30mA	0.02883	0.02771	40.00000
sg13g2_IOPadTriOut4mA	0.02883	0.02771	12.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadTriOut16mA	150.02200	2578.01060	5696.05000
sg13g2_IOPadTriOut30mA	266.97800	2679.28960	5697.69000
sg13g2_IOPadTriOut4mA	50.28470	2498.96296	5695.97000

Delay Information

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p->pad (RR)	0.02000	1.00000	1.44190	0.33000	9.00000	2.44854	2.50000	20.00000	3.65974
	c2p_en->pad (FR)	0.02000	1.00000	1.03676	0.33000	9.00000	1.08721	2.50000	20.00000	1.25410
	c2p_en->pad (RR)	0.02000	1.00000	1.44332	0.33000	9.00000	2.50613	2.50000	20.00000	3.80080
sg13g2_IOPadTriOut30mA	c2p->pad (RR)	0.02000	2.00000	1.79086	0.33000	18.00000	3.01288	2.50000	40.00000	4.32554
	c2p_en->pad (FR)	0.02000	2.00000	1.22935	0.33000	18.00000	1.28079	2.50000	40.00000	1.44222
	c2p_en->pad (RR)	0.02000	2.00000	1.78818	0.33000	18.00000	3.08268	2.50000	40.00000	4.48540
sg13g2_IOPadTriOut4mA	c2p->pad (RR)	0.02000	1.00000	1.45674	0.33000	5.00000	3.22059	2.50000	12.00000	6.25900
	c2p_en->pad (FR)	0.02000	1.00000	0.84943	0.33000	5.00000	0.90025	2.50000	12.00000	1.06083
	c2p_en->pad (RR)	0.02000	1.00000	1.47022	0.33000	5.00000	3.30927	2.50000	12.00000	6.48935

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p->pad (FF)	0.02000	1.00000	1.03048	0.33000	9.00000	1.94212	2.50000	20.00000	3.36021
	c2p_en->pad (FF)	0.02000	1.00000	1.02052	0.33000	9.00000	1.09036	2.50000	20.00000	1.32404
	c2p_en->pad (RF)	0.02000	1.00000	0.86296	0.33000	9.00000	1.87685	2.50000	20.00000	3.15247
sg13g2_IOPadTriOut30mA	c2p->pad (FF)	0.02000	2.00000	1.39776	0.33000	18.00000	2.42540	2.50000	40.00000	3.87735
	c2p_en->pad (FF)	0.02000	2.00000	1.61570	0.33000	18.00000	1.68017	2.50000	40.00000	1.87955
	c2p_en->pad (RF)	0.02000	2.00000	1.03799	0.33000	18.00000	2.19846	2.50000	40.00000	3.55875
sg13g2_IOPadTriOut4mA	c2p->pad (FF)	0.02000	1.00000	1.08701	0.33000	5.00000	2.85624	2.50000	12.00000	6.12395
	c2p_en->pad (FF)	0.02000	1.00000	0.52541	0.33000	5.00000	0.59267	2.50000	12.00000	0.82246
	c2p_en->pad (RF)	0.02000	1.00000	1.11012	0.33000	5.00000	2.99771	2.50000	12.00000	6.25498

Power Information

Internal switching power(pJ) to pad rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p	0.02000	1.00000	6.10605	0.33000	9.00000	5.93894	2.50000	20.00000	5.60073
	c2p	0.02000	1.00000	0.07966	0.33000	9.00000	0.10127	2.50000	20.00000	0.43402
	c2p_en	0.02000	1.00000	5.76392	0.33000	9.00000	8.23063	2.50000	20.00000	11.50290
	c2p_en	0.02000	1.00000	0.03338	0.33000	9.00000	0.06438	2.50000	20.00000	0.41936
sg13g2_IOPadTriOut30mA	c2p	0.02000	2.00000	9.76917	0.33000	18.00000	9.30065	2.50000	40.00000	8.85093
	c2p	0.02000	2.00000	0.07967	0.33000	18.00000	0.10128	2.50000	40.00000	0.43411
	c2p_en	0.02000	2.00000	9.19248	0.33000	18.00000	14.52310	2.50000	40.00000	21.23290
	c2p_en	0.02000	2.00000	0.03339	0.33000	18.00000	0.06439	2.50000	40.00000	0.41944
sg13g2_IOPadTriOut4mA	c2p	0.02000	1.00000	3.18536	0.33000	5.00000	3.06128	2.50000	12.00000	2.85945
	c2p	0.02000	1.00000	0.07969	0.33000	5.00000	0.10131	2.50000	12.00000	0.43372
	c2p_en	0.02000	1.00000	2.97232	0.33000	5.00000	3.77527	2.50000	12.00000	5.18268
	c2p_en	0.02000	1.00000	0.03342	0.33000	5.00000	0.06441	2.50000	12.00000	0.42171

Internal switching power(pJ) to pad falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p	0.02000	1.00000	14.45800	0.33000	9.00000	9.15686	2.50000	20.00000	7.21684
	c2p	0.02000	1.00000	0.19078	0.33000	9.00000	0.21601	2.50000	20.00000	0.54912
	c2p_en	0.02000	1.00000	3.18075	0.33000	9.00000	3.18137	2.50000	20.00000	3.18188
	c2p_en	0.02000	1.00000	0.05520	0.33000	9.00000	0.07150	2.50000	20.00000	0.25743
sg13g2_IOPadTriOut30mA	c2p	0.02000	2.00000	52.67230	0.33000	18.00000	31.81810	2.50000	40.00000	22.82060
	c2p	0.02000	2.00000	0.19080	0.33000	18.00000	0.21604	2.50000	40.00000	0.54944
	c2p_en	0.02000	2.00000	5.00667	0.33000	18.00000	5.01063	2.50000	40.00000	5.01109
	c2p_en	0.02000	2.00000	0.05522	0.33000	18.00000	0.07149	2.50000	40.00000	0.26145
sg13g2_IOPadTriOut4mA	c2p	0.02000	1.00000	2.27792	0.33000	5.00000	2.26356	2.50000	12.00000	2.26908
	c2p	0.02000	1.00000	0.19099	0.33000	5.00000	0.21621	2.50000	12.00000	0.54961
	c2p_en	0.02000	1.00000	1.65061	0.33000	5.00000	1.64878	2.50000	12.00000	1.64881
	c2p_en	0.02000	1.00000	0.05537	0.33000	5.00000	0.07170	2.50000	12.00000	0.25941

Passive power(pJ) for c2p rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	-0.02949	0.33000	-0.03088	2.50000	-0.03146
sg13g2_IOPadTriOut30mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	-0.02949	0.33000	-0.03088	2.50000	-0.03146
sg13g2_IOPadTriOut4mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	-0.02949	0.33000	-0.03089	2.50000	-0.03146

Passive power(pJ) for c2p falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	0.03649	0.33000	0.03620	2.50000	0.03596
sg13g2_IOPadTriOut30mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	0.03661	0.33000	0.03620	2.50000	0.03596
sg13g2_IOPadTriOut4mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	0.03650	0.33000	0.03619	2.50000	0.03596

Passive power(pJ) for c2p rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	-0.02949	0.33000	-0.03088	2.50000	-0.03146
sg13g2_IOPadTriOut30mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	-0.02949	0.33000	-0.03088	2.50000	-0.03146
sg13g2_IOPadTriOut4mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	-0.02949	0.33000	-0.03089	2.50000	-0.03146

Passive power(pJ) for c2p falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	0.03649	0.33000	0.03620	2.50000	0.03596
sg13g2_IOPadTriOut30mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	0.03661	0.33000	0.03620	2.50000	0.03596
sg13g2_IOPadTriOut4mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	0.03650	0.33000	0.03619	2.50000	0.03596